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Tectonic deformation of the Indochina Peninsula recorded in the Mesozoic palaeomagnetic results

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SUMMARY

In order to describe features of tectonic deformation in the Indochina Peninsula, Early Jurassic to Early Cretaceous red sandstones were sampled at three localities in the Shan-Thai and Indochina blocks. Stepwise thermal treatment of most samples revealed the presence of characteristic remanent magnetization, which is generally unblocked by 680 °C. This component from Phong Saly (21.6°N, 101.9°E) and Borikhanxay (18.5°N, 103.8°E) localities yield positive fold tests with Late Jurassic–Early Cretaceous directions of Dec/Inc = 28.8°/32.1° (k_s = 15.4, α_{95} = 8.8°, N = 22) and Dec/Inc = 42.1°/46.9° (k_s = 20.1, α_{95} = 7.9°, N = 18), respectively. Additionally, a syn-folding mid-Cretaceous characteristic magnetization is observed in the samples of Muang Phin locality (16.5°N, 106.1°E), which gave a mean direction of Dec/Inc = 30.8°/39.9°, k = 102.6, α_{95} = 3.0°, N = 23. This reliable Late Jurassic to Mid-Cretaceous palaeomagnetic directions from three different localities are incorporated into a palaeomagnetic database for Shan-Thai and Indochina blocks. Based on these compilations, tectonic deformation of the Shan-Thai and Indochina blocks is summarized as follows: (1) the Shan-Thai and Indochina blocks experienced a clockwise rotation of about 10° as a composite unit in the early stage of India–Asia collision and (2) following this, the Shan-Thai Block underwent an internal tectonic deformation, whereas the Indochina Block behaved as a rigid tectonic unit during the same period. Comparison of our palaeomagnetic results with seismic tomographic images suggests that the strength of continental lithosphere beneath these blocks played an important role in the process of deformation rather than any other tectonic regime. In contrast to the Shan-Thai Block, an existence of continental roots beneath the Indochina Block prevented its internal deformation.

Key words: Palaeomagnetism applied to tectonics; Continental tectonics: compressional; Asia.

1 INTRODUCTION

It is well documented that the Asian Continent was subjected to large-scale tectonic deformation as a result of continued northward indentation of the Indian subcontinent after their initial collision 50 Ma (e.g. Molnar & Tapponnier 1975; Tapponnier & Molnar 1979; Tapponnier *et al.* 1982; Houseman & England 1986; Rowley 1996; Aitchison *et al.* 2007). Recently, the space geodetic studies have provided new insight in to present-day deformational features of the Asian Continent, which clearly indicate clockwise rotational movement around the eastern Himalayan syntaxes (Wang *et al.* 2001b; Zhang *et al.* 2004; Calais *et al.* 2006). With the help of most recent GPS observation, several dynamic models have been proposed in order to explain large-scale tectonic structure of the Tibetan plateau and its neighbouring terranes (England & Molnar 2005; Flesch *et al.* 2005; Copely & McKenzie 2007; Gan *et al.* 2007; Meade 2007; Vergnolle *et al.* 2007; Wang *et al.* 2008). However, a high magnitude of clockwise rotation observed around eastern Himalayan syntaxes can hardly be justified through these models, which also include a viscous flow of continuously deforming solid with the present-day strain rate data (Vergnolle *et al.* 2007; Wang *et al.* 2008). Discrepancy between different aspects of deformation (particularly between observations and models) is partly ascribable to lack of information about long-term deformational features as

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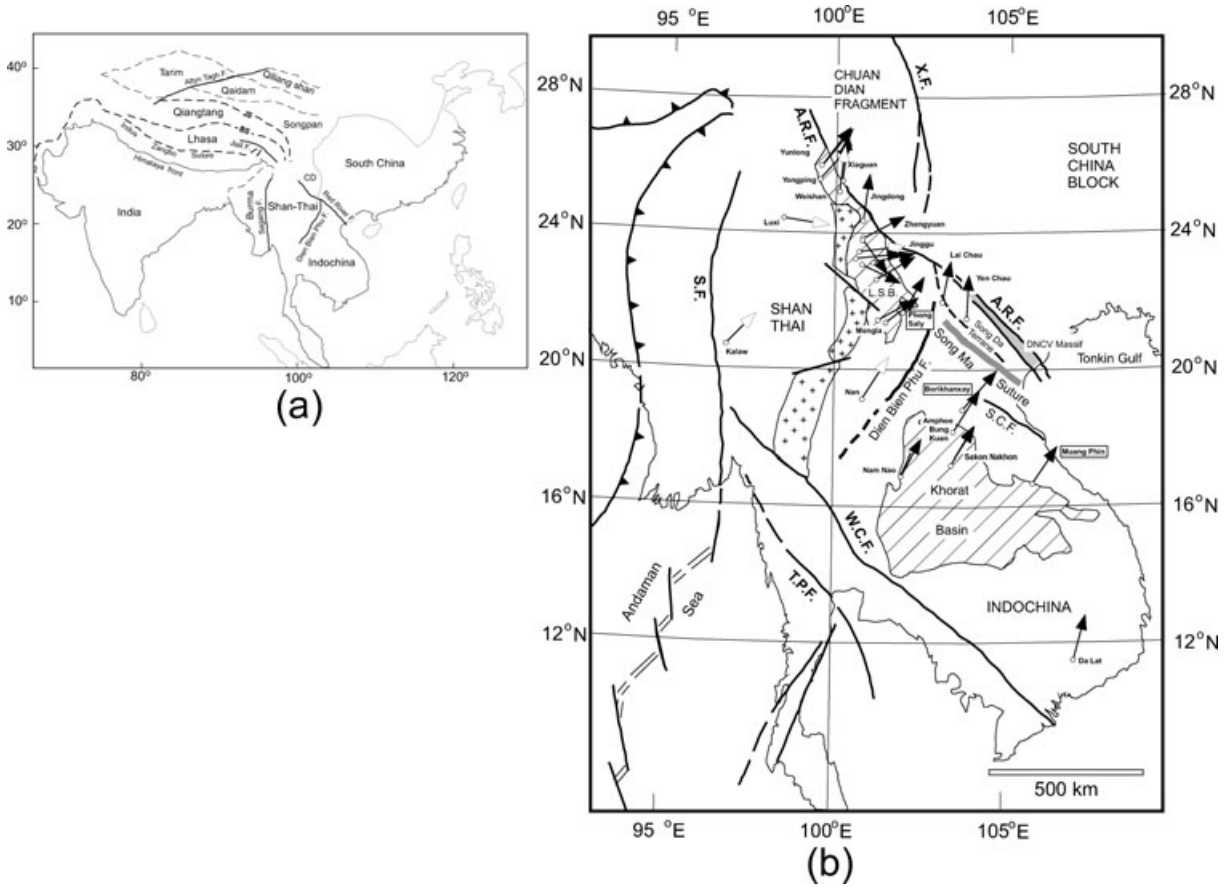


Figure 1. (a) A simplified tectono-geographical map of southeast-Asia showing positions of main sutures and faults. BS: Bangong-Nujiang suture; JS, Jinsha suture; CD: Chuan Dian Fragment. (b) Structural sketch map of the Shan-Thai Block and neighbouring areas (modified from Leloup *et al.* 1995; Lacassin *et al.* 1998 and Shen *et al.* 2005). Shaded areas indicate distribution of the Jurassic to Cretaceous red beds. Names in boxes indicate our study localities. Observed mean declinations at each sampling locality from Shan-Thai and Indochina blocks are indicated by arrows (Jurassic: open arrow, Cretaceous: solid arrow). A.R.F., Ailao Shan-Red River Fault; DNCV Massif, Day Nui Con Voi Massif; L.S.B, Lanping-Simao Basin, S.C.F., Song Ca Fault; S.F., Sagaing Fault,; T.P.F., Three Pagodas Fault; W.C.F., Wang Chao Fault and X.F., Xianshuihe Fault.

well as the fundamental rheological properties from the Asian Continent (Clark & Royden 2000; Replumaz & Tapponnier 2003; Flesch *et al.* 2005; Lev *et al.* 2006; Copely & McKenzie 2007; Thatcher 2007). As mentioned in the literature, knowledge about long-term deformational aspects can help us to understand plate-scale motions of the rigid blocks, while rheological properties can tell us about forces associated with large-scale collision boundary.

Record of the Mesozoic palaeomagnetic declinations can provide us some clue about the post-Cretaceous cumulative rotation of the rigid blocks in a variety of tectonic framework, particularly after the terminal collision of India with Asia. The available palaeomagnetic data from East Asia reveal an occurrence of rotational motion mainly in the Shan-Thai, Indochina and Chuan Dian blocks, which forms part of eastern Himalayan syntaxes (Funahara *et al.* 1992, 1993; Huang & Opdyke 1993; Yang & Besse 1993; Otofuji *et al.* 1998; Sato *et al.* 1999, 2007; Yang *et al.* 2001; Yoshioka *et al.* 2003; Aihara *et al.* 2007; Fig. 1). Further up-gradation of palaeomagnetic record (Yoshioka *et al.* 2003; Aihara *et al.* 2007; Tanaka *et al.* 2008) indicates an occurrence of intrablock deformation, which has been recorded as a clockwise rotation of variable degree in the Shan-Thai and Chuan Dian blocks (Fig. 1). These intrablock deformational features can shed further light on long-term rheological properties of these blocks.

In this paper, we are focusing our attention on intrablock deformational regimes in the Indochina Peninsula, which is a part of eastern Himalayan syntaxes. With the exception of Khorat Plateau (Yang & Besse 1993; Charusiri *et al.* 2006), the available reliable palaeomagnetic data sets from target area is not enough to delineate these complex deformational features in a systematic manner. Keeping in view this demand for further data, we have collected palaeomagnetic samples from Jurassic to mid-Cretaceous red beds at three different areas (including the Phong Saly, Borikhanxay and Muang Phin localities) of the Shan-Thai and Indochina blocks; (Fig. 1b). The purpose of this study is to precisely delineate deformational pattern of the Indochina Peninsula by using an updated palaeomagnetic database.

2 GEOLOGICAL SETTING AND SAMPLING

The Indochina Peninsula, which includes the Shan-Thai and Indochina blocks, is an excellent example of the Cenozoic evolution in southeastern Asia (e.g. Leloup *et al.* 1995, 2001; Morley 2002). The Shan-Thai Block is situated to the south of eastern Himalaya syntaxes and is separated from the South China Block (SCB) by

NW–SE trending Ailao Shan-Red River Fault system. Towards east, the Indochina Block forms part of the Indochina Peninsula and is separated from the SCB and Shan-Thai Block by the Ailao Shan-Red River Fault system and the NE–SW trending Dien Bien Phu Fault (DBPF), respectively.

Several large-scale red bed basins of the Mesozoic era exist within the Indochina Peninsula (BGMRY 1990; Tien *et al.* 1991; Mantajit *et al.* 2002) (Fig. 1). Almost half of the Shan-Thai Block is occupied by the Lanping-Simao Basin, while the central part of the Indochina Block is formed by the Khorat Basin. In this study, we are focusing on the Upper Jurassic to Lower Cretaceous strata exposed in these two basins. Palaeomagnetic samples were collected at Phong Saly locality (21.6°N, 101.9°E) of eastern Lanping-Simao Basin, the Borikhanxay locality (18.5°N, 103.8°E) of northern Khorat Basin, and the Muang Phin locality (16.5°N, 106.1°E) of eastern Khorat Basin (Fig. 2).

2.1 The Phong Saly locality of the Lanpin-Simao Basin

Here the Mesozoic to Cretaceous continental sediments rest unconformably over pre-Mesozoic strata. In Chinese territory, these sediments are subdivided into four observable formations as follows in the ascending order: the Lower Cretaceous Jinxing Formation, the Middle Cretaceous Nanxing Formation, the Upper Cretaceous Hutousi Formation and the Upper Cretaceous Mankuanhe Formation (BGMRY 1990; Leloup *et al.* 1995). The presence of rich *Lamellibranchiate*, such as the *Estheria*, *Darwinula*, *Gasterpods* and *Sporopollen*, assigns an age of Early Cretaceous to the Jinxing Formation (BGMRY 1990). Across the boundary in Laos, the Upper Jurassic to Lower Cretaceous strata of this basin is described as the Pudendin Formation. This formation with a total thickness of 600–800 m is characterized by red colour continental sediments, which mainly composed of oblique-bedded sandstones, siltstones and interbeded conglomerates and gritstones (Tien *et al.* 1991). These stratified rocks are visibly affected by NNW–SSE trended folding and thrusting (Leloup *et al.* 1995). A discontinuity between Middle to Upper Eocene and Lower Eocene strata (BGMRY 1990) indicates an occurrence of folding related activities during Late Eocene (e.g. Leloup *et al.* 1995). Samples were collected at 22 sites from the Upper Jurassic to Lower Cretaceous Pudendin Formation at Phong Saly Locality, where bedding attitude is dipping towards west between 19° and 49° (Fig. 2a).

2.2 The Borikhanxay and Muang Phin localities in the Khorat Basin

Major part of the Indochina Block is occupied by the Khorat Basin, which is filled by non-marine Mesozoic strata up to 5000 m thick (Mantajit *et al.* 2002). In northern part of the Khorat Basin, the Palaeozoic to Triassic basement is covered by a red bed sequence (The Khorat Group), which in turn is overlain by the Phong Hong Group. Fossils related evidences assign an age of Early Jurassic–Early Cretaceous to the Khorat Group and Late Cretaceous to the Phong Hong Group. On the Laos side, the Jurassic–Cretaceous strata are divided into three units in ascending order: the Lower-Middle Jurassic Tholam Formation, the Upper Jurassic–Lower Cretaceous Champa Formation, and the Upper Cretaceous Donghen Formation (Tien *et al.* 1991). The Khorat Group in northern Khorat Plateau is significantly affected by regional tilting together with compressive folding and reverse faulting, whereas tilting of the Phong Hong Group rarely exceeds a few degrees. According to Smith *et al.* (1996) observations, rocks in the Khorat Basin have been affected by regional scale compressive tectonics activities in the mid-Cretaceous (Aptian–Cenomanian). For this study, we are focusing our attention on two areas in the Khorat Basin; that is, the Borikhanxay locality near its northern margin and the Muang Phin locality along its eastern margin. At Borikhanxay locality, samples were collected from the Upper Jurassic to Lower Cretaceous Champa Formation at 18 sites, where a bedding attitude of 18°–77° towards northeast or southwest was observed (Fig. 2b). Samples at Muang Phin locality were collected from the Lower-Middle Jurassic Tholam Formation at 30 sites. As evident from Fig. 2(c), bedding attitude at this locality shows a variation in strike as well as in dipping angle (from 3.5° to 41°).

Three to ten block samples, oriented with magnetic compass, were collected from each site. The present-day declination at each sampling site was evaluated from the International Geomagnetic Reference Field (Macmillan & Maus 2005).

3 PALAEOMAGNETISM

3.1 Laboratory procedures

One or more specimens, 25 mm in diameter and 22 mm in height, were prepared from each block sample in the palaeomagnetic laboratory of the Kobe University. Remanent magnetization of each specimen was measured using a 2G Enterprises cryogenic magnetometer. Stepwise thermal demagnetization was carried out up to 690 °C using a Natsuhara TDS-1 thermal demagnetizer, where a residual field in the furnace was less than 5 nT. In order to monitor possible chemical changes in the magnetic mineralogy during thermal treatment, magnetic susceptibility was measured after each heating step using a Bartington MS2 susceptibility metre. Results from most specimens showed no significant change in susceptibility during heating procedure, implying that no thermal alteration of magnetic minerals has occurred. Magnetic behaviour of each specimen after complete demagnetization was plotted on the Zijderveld diagrams (Zijderveld 1967). Principal component analysis (Kirschvink 1980) was used to determine directional trend of different magnetization components. Site-mean directions were calculated by using the Fisherian statistics (Fisher 1953).

3.2 Demagnetization results and mean directions

3.2.1 The Phong Saly locality

Measurements of specimens from this locality (Upper Jurassic to Lower Cretaceous Pudendin Formation) revealed their initial natural remanent magnetization (NRM) intensities between 0.19 and 25 mA m^{-1}. Generally, two components of magnetization are isolated. After the removal of low-temperature component by 150–350 °C, the high-temperature component then appeared and linearly decayed toward origin between 670 and 690 °C (Figs 3a–c).

The low-temperature component, which we observed in all 22 sites, gave a formation mean direction of Dec/Inc = 10.4°/26.3°, kg = 13.5, α_{95} = 8.8° (N = 22) in geographic coordinate. This direction is nearly parallel to the IGRF direction (D = 359.3°, I = 30.3°) in the study area, which strongly advocates a viscous remanent magnetization (VRM) origin for this component.

However, the high-temperature component is found to be carrying both normal and reversed polarities, where 17 out of 22 sites revealed normal polarity and the remaining five sites gave a reversed

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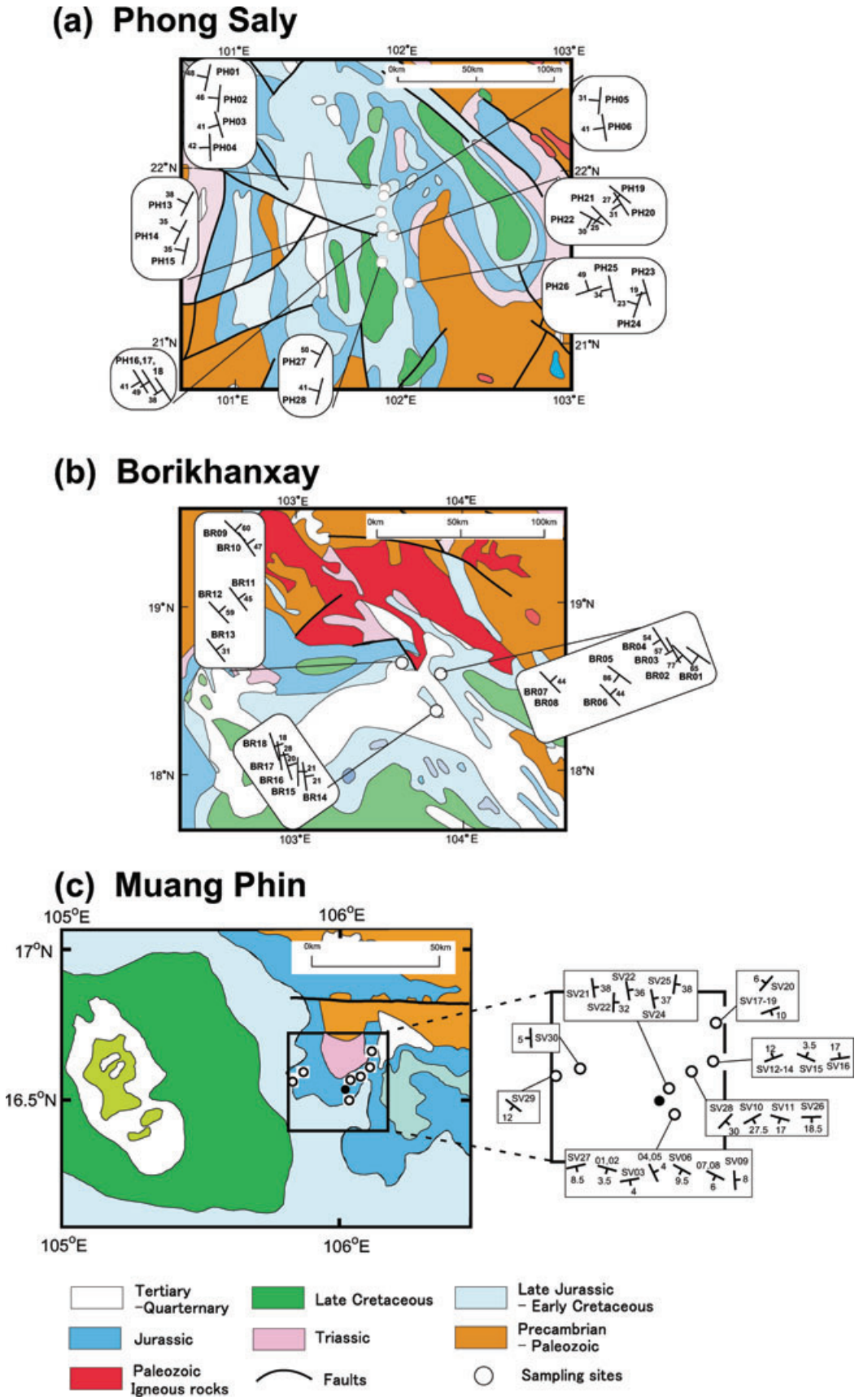


Figure 2. Geological map of the study area (after BGMRY (1990)): (a) the Phong Saly locality, (b) the Borikhanxay locality and (c) the Muang Phin locality. Sampling sites are shown by closed circles. Strike and dip of strata for each sampling site are shown in the insets; P, Paleozoic; T, Triassic; J, Jurassic; K, Cretaceous; E, Eocene and N, Neogene.

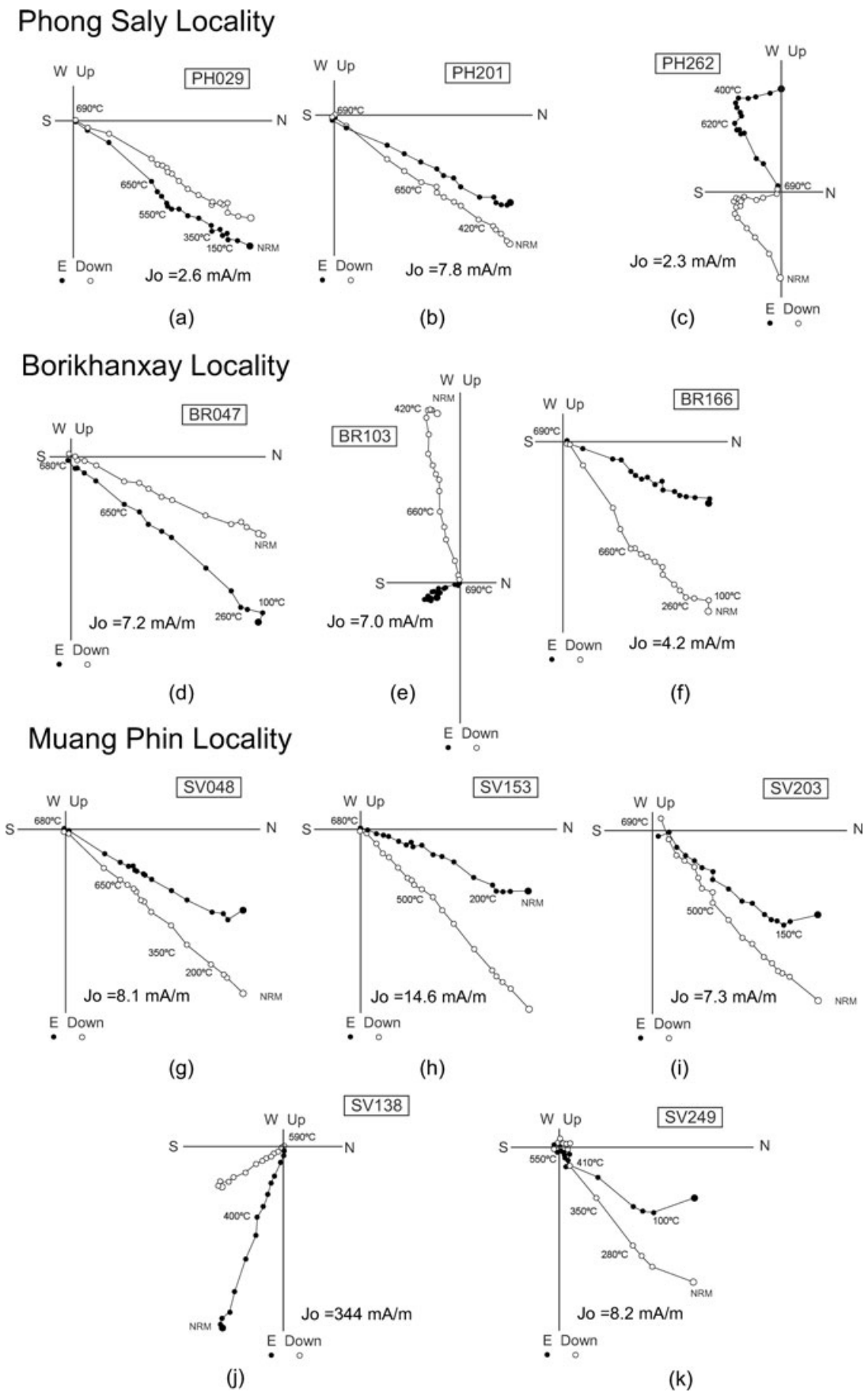


Figure 3. Vector end-point diagrams for representative specimens from Phong Saly (a)–(c), Borikhanxay (d)–(f) and Muang Phin (g)–(k) localities after thermal demagnetization experiments (in geographic coordinates). Solid (open) symbols are projections on to horizontal (vertical) plane.

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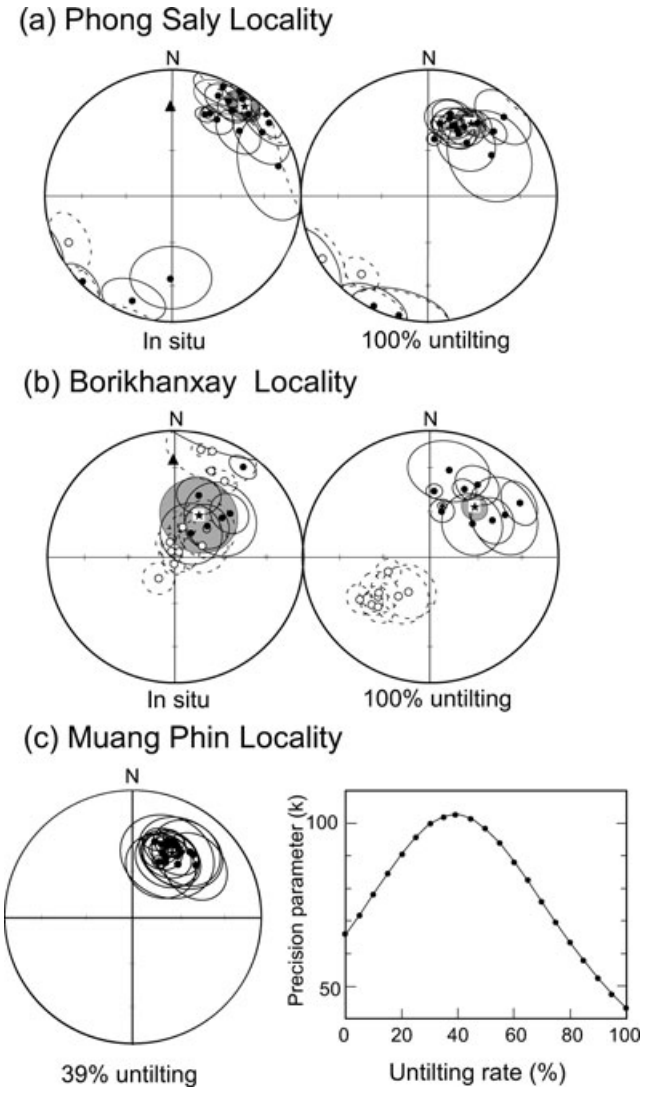


Figure 4. Equal-area projections of the site mean directions (circles) for high-temperature component with 95 per cent confidence limit before and after tilt correction for (a) Phong Saly, and (b) Borikhanxay localities (c) Site mean directions of the Muang Phin locality at 39 per cent untilting together with progressive untilting of the formation mean directions at 5 per cent increment. The precision parameter (k) reaches to its maximum value at 39 per cent untilting. The formation mean direction of each locality is shown by star with 95 per cent circle of confidence (shaded). Solid triangle represents the present-day direction of the Earth's magnetic field. Solid and open symbols correspond to projection on lower and upper hemisphere, respectively.

polarity direction. Due to large dispersion in directional behaviour ($\alpha_{95} > 30^\circ$), data of 3 out of 22 sites (PH03, 06, 27) are discarded for further palaeomagnetic discussion. After flipping the reversed polarity directions into normal, the formation mean direction of 19 sites is calculated as Dec/Inc $= 37.7^\circ/11.2^\circ$ ($k_g = 12.7$, $\alpha_{95} = 9.8^\circ$) in geographic coordinates and Dec/Inc $= 28.8^\circ/32.1^\circ$ ($k_s = 15.4$, $\alpha_{95} = 8.8^\circ$) in stratigraphic coordinates (Fig. 4a).

Although, the Upper Jurassic–Lower Cretaceous strata form a monoclinal structure, the bedding attitude in the study area slightly varies through the stratigraphic sequence. We therefore, applied a second definition (ξ_2) of the McFadden's (1990) fold test to our data of 19 sites. This fold test proved positive at 95 and 99 per cent confidence levels, where a calculated value (ξ_2) is 7.966 in geographic coordinates and 3.973 in stratigraphic coordinates, while critical value ξ_{95} (ξ_{99}) is 5.075 (7.112) at 95 per cent (99 per cent) confidence levels, (Cogne 2003). Furthermore, the direction-correlation (DC) tilt test (Enkin 2003) gave an optimal concentration at 68 ± 43 per cent untilting, which is indistinguishable from that at 100 per cent untilting.

In addition to above-mentioned procedure, we have selected several subsets of our sampling sites for further fold tests. For this purpose only those subsets of sites were selected where a difference in bedding attitude is largest between them. A subset of six sites from the central part of Phong Saly locality (PH13, PH14, 15, PH19, PH21 and PH22) shows a maximum value of k at 98 per cent unfolding, which suggests a positive fold test (McFadden 1990) at 95 per cent confidence level. From this test, a calculated value (ξ_2) of 3.879 is obtained in geographic coordinates and 1.505 in stratigraphic coordinates, while a critical value (ξ_c) of 2.962 is obtained at 95 per cent confidence level. We therefore recognize that the formation mean direction of 19 sites at 100 per cent untilting represents a characteristic remanent magnetization (ChRM) for the Upper Jurassic to Lower Cretaceous Pudendin Formation.

3.2.2 The Borikhanxay locality

Treatment of samples from the Upper Jurassic to Lower Cretaceous Champa Formation revealed their initial NRM intensities between 1.2 and 42 mA m^{-1}. Similar to the Phong Saly locality, two components of magnetization are isolated as a result of thermal treatment. After the removal of low-temperature component between 200 and 300 °C, the high-temperature component followed a linear decay toward origin and unblocked between 670 and 690 °C (Figs 3d–f).

The low-temperature component is observed in all 18 sites. The *in situ* formation mean direction for this component is Dec/Inc $= 16.5^\circ/23.1^\circ$, $k_g = 4.4$, $\alpha_{95} = 18.6^\circ$ ($N = 18$), which is almost identical to the IGRF direction ($D = 359.4^\circ$, $I = 24.3^\circ$) in the study area. This type of behaviour indicates a VRM origin for this component.

Similar to Phong Saly locality, the high-temperature component from this locality revealed both normal and reversed polarity directions, where 10 out of 18 sites are of normal polarity and the remaining 8 of reversed polarity. By flipping the reversed polarity directions into normal state resulted in a formation mean direction of Dec/Inc $= 30.4^\circ/58.5^\circ$ ($k_g = 3.0$, $\alpha_{95} = 24.1^\circ$) in geographic coordinates and Dec/Inc $= 42.1^\circ/46.9^\circ$ ($k_s = 20.1$, $\alpha_{95} = 7.9^\circ$) in stratigraphic coordinates (Fig. 4b).

Using the McFadden's (1990) fold test, a calculated value (ξ_1) of 10.298 is obtained in geographic coordinates and 3.835 in stratigraphic coordinates, while a critical value (ξ_c) of 6.919 is obtained at 95 per cent confidence level. The application of DC tilt test (Enkin 2000) revealed a positive result at 94 ± 14 per cent untilting. The tilt-corrected normal (Dec/Inc $= 35.6^\circ/44.3^\circ$, $k_g = 15.0$, $\alpha_{95} = 12.9^\circ$, $N = 10$) and reversed (Dec/Inc $= 230.7^\circ/-49.6^\circ$, $k_s = 43.4$, $\alpha_{95} = 8.5^\circ$, $N = 8$) polarities mean directions show a positive reversal test (McFadden & McElhinny 1990) with classification 'C'. An angular diistance (γ)) obtained by this procedure is 11.5°, which is less than the critical angle of $\gamma_c = 15.6^\circ$. We thus recognize that the formation mean direction at 100 per cent untilting represents a ChRM for the Upper Jurassic to Lower Cretaceous Champa Formation.

3.2.3 The muang phin locality

With the exception of site SV13, thermal treatment of specimens from the Lower-Middle Jurassic Tholam Formation revealed their initial NRM intensities between 2.2 and 46.2 mA m^{-1}. The NRMs of site SV13, which carry relatively high intensity (between 337 and 344 mA m^{-1}), showed a single component behaviour with unblocking temperature of 580 °C (Fig. 3j). The NRM of this site was probably acquired as a result of lightning strikes. On the other hand, we have found two to three components of magnetization from other sites (Figs 3g–k). In addition to low-temperature component by 200 or 300 °C, a high-temperature component appeared and eventually followed a linear decay toward origin between 670 and 690 °C in 24 sites (Figs 3g–i). In the remaining five sites (SV19, SV21, SV22, SV23 and SV24), after the separation of low-temperature component by 200 °C, an intermediate-temperature component appeared between 200 and 570 °C from but a high-temperature component was not isolated by the principal component analysis (Fig. 3k).

The low-temperature component is obtained from 12 sites, which gives an *in situ* formation mean direction of Dec/Inc = 358.9°/33.3°, kg = 35.4, $\alpha_{95} = 13.0°$ ($N = 12$). Similar to other localities, a mean direction for this component is also identical to that of the geocentric axial dipole field in the study area ($D = 0$, $I = 30.6°$), and is thus attributed to recently acquired VRM.

The high-temperature component with exclusive normal polarity direction is identified in 25 sites. Because of large dispersion in remanent directions ($\alpha_{95} > 30°$), results of seven sites (denoted by asterisk in Table 1) are excluded from the formation mean calculation. The *in situ* formation mean direction for the remaining 18 sites is calculated as Dec/Inc = 28.2°/39.2° ($k_g = 73.5$, $\alpha_{95} = 4.1°$). During the progressive unfolding procedure, these site-mean directions gradually approached each others and eventually merged into tight clustering at middle level. Further unfolding up to 100 per cent level, however, produces a dispersion in directional behaviour (Dec/Inc = 31.5°/40.4°; $k_s = 62.0$, $\alpha_{95} = 4.5°$).

After the application of DC tilt test (Enkin 2003) to this high-temperature component from 18 sites, an optimal concentration is achieved at 43 ± 24 per cent untilting. Results from DC tilt test and the progressive unfolding procedure thus indicate a syn-folding origin for this component. The formation mean direction at 43 per cent untilting is calculated as Dec/Inc = 29.6°/39.9°, $k = 102.0$, $\alpha_{95} = 3.4°$, $N = 18$.

Because of inconclusive DC-tilt test (where an optimal concentration is obtained at 54 ± 75 per cent untilting), syn-folding origin is assigned to intermediate-temperature component (5 sites) as well. Mean direction obtained from these five sites at 54 per cent untilting (Dec/Inc = 38.9°/37.2°, $k = 137.7$, $\alpha_{95} = 6.5°$, $N = 5$) is found to be sub parallel to that of high-temperature component. Due to this directional similarity between the high and intermediate-temperature components, the DC-tilt test is then applied to characteristic directions from all 23 sites. A syn-folding origin is assigned because of the optimal concentration at 39 ± 18 per cent (Fig. 4c). The formation mean direction from 23 sites at 39 per cent untilting (Dec/Inc = 30.8°/39.9°, $k = 102.6$, $\alpha_{95} = 3.0°$, $N = 23$) is thus recognized as a ChRM for Tholam Formation at Muang Phin locality (Fig. 4c).

4 ROCK MAGNETISM

In order to identify magnetic minerals in the studied samples, progressive acquisition of isothermal remanent magnetization (IRM) up to 2.7T and thermal demagnetization of the composite IRMs (Lowrie 1990) was performed by using the 2G pulse magnetizer. For composite IRMs, hard, medium and soft components were treated in DC fields of 2.7T, 0.4T and 0.12T, respectively.

Rock magnetic experiments on the red beds of all three localities suggest the presence of hematite as a dominant magnetic carrier (Fig. 5). IRM acquisition curves indicate that no saturation is achieved up to a maximum field of 2.7T (with the coercivity of remanent magnetization is 400–600 mT), indicating a dominancy of high-coercivity minerals, such as hematite. The presence of hematite is further confirmed by thermal demagnetization of the composite IRMs, where an unblocking temperature of around 680 °C is obtained for hard component.

An unblocking temperature of 580 °C is observed in the soft component of two specimens from Muang Phin Locality (Fig. 5c). The IRM acquisition curves also indicate an initial steep increase with upward convex shape up to100 mT, signifying the presence of magnetite.

Combined with thermal demagnetization results (NRM), the intermediate temperature component obtained from the Muang Phin locality is most likely carried by magnetite, whereas the high-temperature component in all three localities is carried by hematite. Occurrence of magnetite in the red beds may be linked to secondary mineralogical changes as a result of folding related activities in the Muang Phin area.

5 DISCUSSION

New reliable palaeomagnetic results have been obtained from the Lower Jurassic to Lower Cretaceous red beds collected at three different localities in the Laos. Pre-folding origin of the data sets from Phong Saly and Borikhanxay localities is ascertained through positive fold tests. Easterly deflected palaeomagnetic directions are obtained from these two localities. The Upper Jurassic to Lower Cretaceous Pudendin Formation from the Lanpin-Simao Basin (Phong Saly locality) gave a mean direction of Dec/Inc = 28.8°/32.1°, k_s = 15.4, $\alpha_{95} = 8.8°$, $N = 19$, while the Upper Jurassic to Lower Cretaceous Champa Formation from the Khorat Basin (Borikhanxay locality) revealed a mean direction of Dec/Inc = 42.1°/46.9°, k_s = 20.1, $\alpha_{95} = 7.9°$, $N = 18$).

The high and intermediate-temperature components obtained from Lower-Middle Jurassic rocks of the Muang Phin locality (Dec/Inc = 30.8°/39.9°, $k = 102.6$, $\alpha_{95} = 3.0°$, $N = 23$) is, however, recognized as syn-folding in origin. The folding episode in the Khorat Group has been dated at Mid-Cretaceous (Smith *et al.* 1996), which we consider to be responsible for the acquisition of ChRM in the red beds of Muang Phin locality. The ChRM directions obtained from all three localities are incorporated into a single database for Indochina and Shan-Thai blocks (Table 2).

The present palaeomagnetic study reproduces easterly deflected palaeomagnetic declinations, which have a fair compatibility with those previously reported from the Shan-Thai and Indochina blocks. From the Lanpin-Simao Basin of central Shan-Thai Block (Fig. 1), the Cretaceous palaeomagnetic directions have been reported from the Nanxin Formation (Huang & Opdyke 1993 and Tanaka *et al.* 2008). These directions are $D = 60.8°$, $I = 37.8$, $\alpha_{95} = 7.6°$ for Mengla locality and $D = 51.2°$, $I = 46.4°$, $\alpha_{95} = 5.6°$ for South Mengla locality. Although, the magnitude of easterly deflected declination (Dec = 29.8°) from Phong Saly locality (this study) is relatively smaller than those previously reported, an eastward-deflected direction characterizes the trend of palaeomagnetic directions in the Jurassic to Cretaceous rocks of the Lanpin-Simao Basin, the central part of the Shan-Thai Block.

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Table 1. Palaeomagnetic results of the red bed samples collected from Lower Jurassic to Lower Cretaceous formations at (a) Phong Saly, (b) Borikhanxay and (c) Mue Phin localities.

(a) Site and formation mean directions from the Phong Saly palaeomagnetic locality

Site	Locality				Polarity	*n*/*N*	*In situ*		Tilt-corrected		*k*	α_{95}
	Lat. (°N)	Long. (°E)	(°)				Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)		(°)
The Upper Jurassic–Lower Cretaceous Pudendin Formation [High-temperature component]												
PH01	21.90	101.89	192	48	Reversed	5/8	200.7	11.7	206.5	1.4	22.5	16.5
PH02	21.90	101.89	186	46	Normal	9/9	43.9	17.3	20.4	38.9	110.7	4.9
PH03*	21.89	101.89	163	41	Normal	3/8	45.6	0.1	38.5	35.6	10.4	40.4
PH04	21.89	101.89	178	42	Normal	7/8	53.8	15.4	35.4	46.9	11.9	18.2
PH05	21.86	101.89	184	31	Normal	6/8	52.1	6.0	45.2	28.1	67.2	8.2
PH06*	21.85	101.89	171	41	Normal	5/8	67.3	14.8	62.2	44.6	4.4	41.1
PH13	21.77	101.89	207	38	Normal	8/8	73.7	16.2	56.9	40.5	4.8	28.1
PH14	21.76	101.87	207	35	Normal	9/9	48.9	2.7	43.4	14.5	10.2	16.9
PH15	21.76	101.87	192	35	Normal	4/10	45.1	28.0	21.6	41.4	45.3	13.8
PH16	21.67	101.88	148	41	Normal	4/8	35.4	7.5	25.7	44.4	80.5	10.3
PH17	21.67	101.88	147	49	Reversed	5/8	181.7	34.4	194.2	0.9	12.2	22.8
PH18	21.67	101.88	146	38	Normal	8/10	24.4	5.3	15.5	36.5	16.5	14.1
PH19	21.63	101.94	129	27	Normal	8/8	24.0	14.4	19.8	40.3	21	12.4
PH20	21.62	101.94	147	31	Normal	8/8	24.1	28.8	5.8	52.3	158.1	4.4
PH21	21.62	101.94	135	25	Normal	4/8	30.1	14.0	26.6	38.1	52.4	12.8
PH22	21.62	101.94	118	30	Normal	7/8	32.6	1.6	33.4	31.5	47.3	8.9
PH23	21.35	102.05	164	19	Normal	9/9	21.2	31.1	9.2	41.1	92.5	5.4
PH24	21.35	102.05	196	23	Normal	8/8	29.2	30.7	14.7	33.1	32.0	9.9
PH25	21.35	102.04	167	34	Normal	7/9	34.1	18.9	18.4	41	54.8	8.2
PH26	21.35	102.03	252	49	Reversed	8/8	245.9	−11.8	239.2	−3.1	13.9	15.4
PH27*	21.48	101.87	206	50	Reversed	4/8	226.8	−5.9	215.1	−19.6	7.4	36.2
PH28	21.46	101.87	194	41	Reversed	6/8	226.6	1.1	220.5	−19.8	41.4	10.5
Normal mean direction												
0 per cent	21.6	101.9				15/17	38.3	10.6			22.2	8.3
100 per cent						15/17			26.7	38.8	32.9	6.8
Reversed mean direction												
0 per cent	21.6	101.9				4/5	215.2	9.8			6.1	40.7
100 per cent						4/5			215.0	−5.3	14.1	25.4
Formation mean direction												
0 per cent	21.6	101.9				19/22	37.7	11.2			12.7	9.8
100 per cent						19/22			28.8	32.1	15.4	8.8

(b) Site and formation mean directions from the Borikhanxay palaeomagnetic locality

Site	Locality				Polarity	n/N	*In situ*		Tilt-corrected		*k*	α_{95}
	Lat. (°N)	Long. (°E)	(°)				Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)		(°)
The Upper Jurassic-Lower Cretaceous Champa Formation [High-temperature component]												
BR01	18.50	103.84	125	65	Normal	7/7	13.7	−13.0	3.8	46.5	157.7	4.8
BR02	18.50	103.84	134	**77**	Normal	7/7	32.9	−28.9	29.8	46.7	75.6	7.0
BR03	18.50	103.84	151	57	Normal	5/8	19.5	−11.7	12.4	30.1	12.3	22.8
BR04	18.50	103.84	147	54	Normal	7/8	36.7	−11.7	14.6	59.3	64.2	7.6
BR05	18.50	103.83	127	86	Normal	8/8	22.8	−26.7	13.6	56.2	591.1	2.3
BR06	18.49	103.83	317	44	Reversed	8/8	15.0	−70.1	249.6	−61.2	48.4	8.0
BR07	18.49	103.83	316	44	Reversed	8/8	328.4	−85.9	232.3	−46.7	75.1	6.4
BR08	18.49	103.83	316	44	Reversed	5/8	67.8	−71.2	211.5	−62.8	20.7	17.2
BR09	18.57	103.74	314	60	Reversed	8/8	48.4	−62.8	218.5	−56.9	9.9	18.5
BR10	18.57	103.74	325	47	Reversed	8/8	183.9	−85.1	230.4	−39.8	63.7	7.0
BR11	18.57	103.74	323	45	Reversed	8/9	42.2	−86.0	234.6	−48.9	19.4	12.9
BR12	18.57	103.74	317	59	Reversed	8/8	343.7	−80.1	238.2	−34.9	40.3	8.8
BR13	18.57	103.74	322	31	Reversed	7/9	216.1	−72.5	225.9	−41.9	36.4	10.1
BR14	18.32	103.85	352	21	Normal	7/8	46.4	−61.6	59.9	43.1	18.8	14.3
BR15	18.32	103.85	359	21	Normal	8/8	49.9	−50.8	60.7	33.3	6.6	23.4
BR16	18.32	103.85	343	20	Normal	7/8	21.3	−46.5	33.3	32.5	22.1	13.1
BR17	18.32	103.85	351	28	Normal	5/8	51.2	−44.2	59.0	18.9	38.6	12.5
BR18	18.32	103.85	341	18	Normal	4/8	35.3	−71.7	52.3	55.5	22.9	19.6
Normal mean direction												
0 per cent	18.5	103.8				10/10	31.0	−21.3			4.2	26.6
100 per cent						10/10			35.6	44.3	15.0	12.9

Table 1. (*Continued.*)

(b) Site and formation mean directions from the Borikhanxay palaeomagnetic locality

Site	Locality			Polarity	n/N	*In situ*		Tilt-corrected		k	α_{95}
	Lat. (°N)	Long. (°E)	(°)			Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)		(°)
Reversed mean direction											
0 per cent	18.5	103.8			8/8	35.8	−83.6			28.6	10.5
100 per cent					8/8			230.7	−49.6	43.4	8.5
Formation mean direction											
0 per cent	18.5	103.8			18/18	30.4	58.5			3.0	24.1
100 per cent					18/18			42.1	46.9	20.1	7.9

(c) Site and formation mean directions from the Muang Phin palaeomagnetic locality

Site	Locality				Polarity	3/3	*In situ*		Tilt-corrected		k	α_{95}
	Lat. (°N)	Long. (°E)	(°)				Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)		(°)
The Lower-Middle Jurassic Tholam Formation [High-temperature component]												
SV01	16.48	106.03	105.3	3.5	Normal	3/3	24.8	48.3	25.7	51.8	989.6	3.9
SV02	16.48	106.03	105.3	3.5	Normal	3/3	31.1	46.6	32.4	49.9	879.5	4.2
SV03	16.48	106.03	80.6	5.7	Normal	3/3	26.1	40.3	29.3	45.0	193.5	8.9
SV04	16.49	106.03	334.5	6.2	Normal	3/3	31.3	36.6	33.4	31.4	321.2	6.9
SV05	16.49	106.03	334.5	6.2	Normal	3/3	23.8	35.9	26.4	31.0	955.0	4.0
SV06	16.49	106.03	120.6	9.0	Normal	3/3	26.5	31.8	26.0	40.8	166.1	9.6
SV07	16.49	106.03	116.3	5.7	Normal	3/3	27.2	33.9	27.5	39.6	307.8	7.0
SV08	16.49	106.03	116.3	5.7	Normal	3/3	31.0	32.2	31.4	37.7	312.4	7.0
SV09	16.49	106.03	355.3	7.5	Normal	3/3	26.4	36.7	30.9	32.6	175.0	9.3
SV10	16.58	106.20	63.3	27.5	Normal	4/4	20.5	35.0	42.4	49.8	92.6	9.6
SV11	16.58	106.11	106.9	18.4	Normal	3/3	38.8	27.1	44.4	43.8	33.1	21.8
SV12*	16.58	106.11	249.1	11.3	Normal	3/3	45.3	41.7	37.2	36.5	9.4	42.8
SV13*	16.58	106.11	249.1	11.3	Normal	3/3	64.1	24.6	59.1	23.2	3.4	80.6
SV14	16.58	106.11	249.1	11.3	Normal	3/3	38.8	42.0	31.2	35.6	93.3	12.8
SV15	16.58	106.11	292.8	12.8	Normal	4/4	22.0	43.1	22.0	30.4	1662.7	2.3
SV16	16.58	106.11	261.3	16.5	Normal	4/4	28.4	48.3	20.2	34.6	60.9	11.9
SV17*	16.65	106.12	70.5	10.1	Normal	3/3	5.3	42.0	10.0	51.0	6.4	53.6
SV18*	16.65	106.12	70.5	10.1	Normal	4/4	2.8	46.7	7.6	55.8	2.3	112.6
SV19	16.65	106.12	70.5	10.1		0/4						
SV20	16.66	106.12	220.3	6.2	Normal	4/4	31.2	42.9	25.7	41.6	35.1	15.7
SV21	16.57	106.07	353.2	36.2		0/3						
SV22	16.57	106.07	2.2	36.6		0/3						
SV23	16.57	106.07	350.2	36.2		0/3						
SV24	16.57	106.07	349.9	38.1		0/3						
SV25	16.57	106.07	9.4	41.2	Normal	4/4	4.0	37.3	32.9	30.3	161.6	7.3
SV26	16.54	106.02	89.7	14.9	Normal	4/4	25.6	46.0	35.8	58.9	19.6	21.3
SV27	16.49	106.30	77.3	7.8	Normal	4/4	48.6	34.5	53.7	37.9	21.0	20.5
SV28*	16.53	106.10	43.8	32.0		3/3	121.0	45.2	124.5	13.6	3.1	87.6
SV29*	16.56	105.80	129.0	12.0	Normal	3/3	22.4	13.6	20.4	10.1	18.3	62.2
SV30*	16.57	105.86	178.7	3.7	Normal	3/3	31.8	46.6	29.6	43.1	3.9	73.5
Formation mean direction (High temperature component)												
0 per cent						18/30	28.2	39.2			73.5	4.1
100 per cent									31.5	40.4	62.0	4.5
43 per cent									29.6	39.9	102.0	3.4
[Intermediate temperature component]												
SV19	16.65	106.12	70.5	10.1	Normal	4/4	33.9	33.4	40.3	39.4	15.5	24.1
SV21	16.57	106.07	353.2	36.2	Normal	3/3	16.8	52.1	43.0	29.4	41.9	19.3
SV22	16.57	106.07	2.2	36.6	Normal	3/3	15.5	47.4	42.8	29.9	20.1	28.3
SV23	16.57	106.07	350.2	36.2	Normal	3/3	36.5	43.1	49.0	13.9	2512.0	2.5
SV24	16.57	106.07	349.9	38.1	Normal	3/3	25.8	54.9	48.8	25.7	65.7	15.3
0 per cent						5/5	26.4	46.5			56.8	10.2
100 per cent									45.0	27.7	68.2	9.3
54 per cent									38.9	37.2	137.7	6.5
Formation mean direction (High and intermediate temperature component)												
0 per cent						23/30	27.9	40.8			65.7	3.8
100 per cent									34.8	37.8	43.2	4.7
39 per cent									30.8	39.9	102.6	3.0

Notes: Magnetization directions for high-temperature component are listed in this table. Lat. and Long., latitude and longitude. N and n are number of samples measured and used for calculation, respectively. Dec. and Inc. are declination and inclination, respectively; k is the Fisherian precision parameter (Fisher 1953); α_{95} is the radius of cone at 95 per cent confidence level about the mean direction. Sites with asterisk (*) are excluded from mean direction calculation. 0, 100 and 39 per cent are parameters in geographic coordinates, strtatigraphic coordinates and at 39 per cent untilting.

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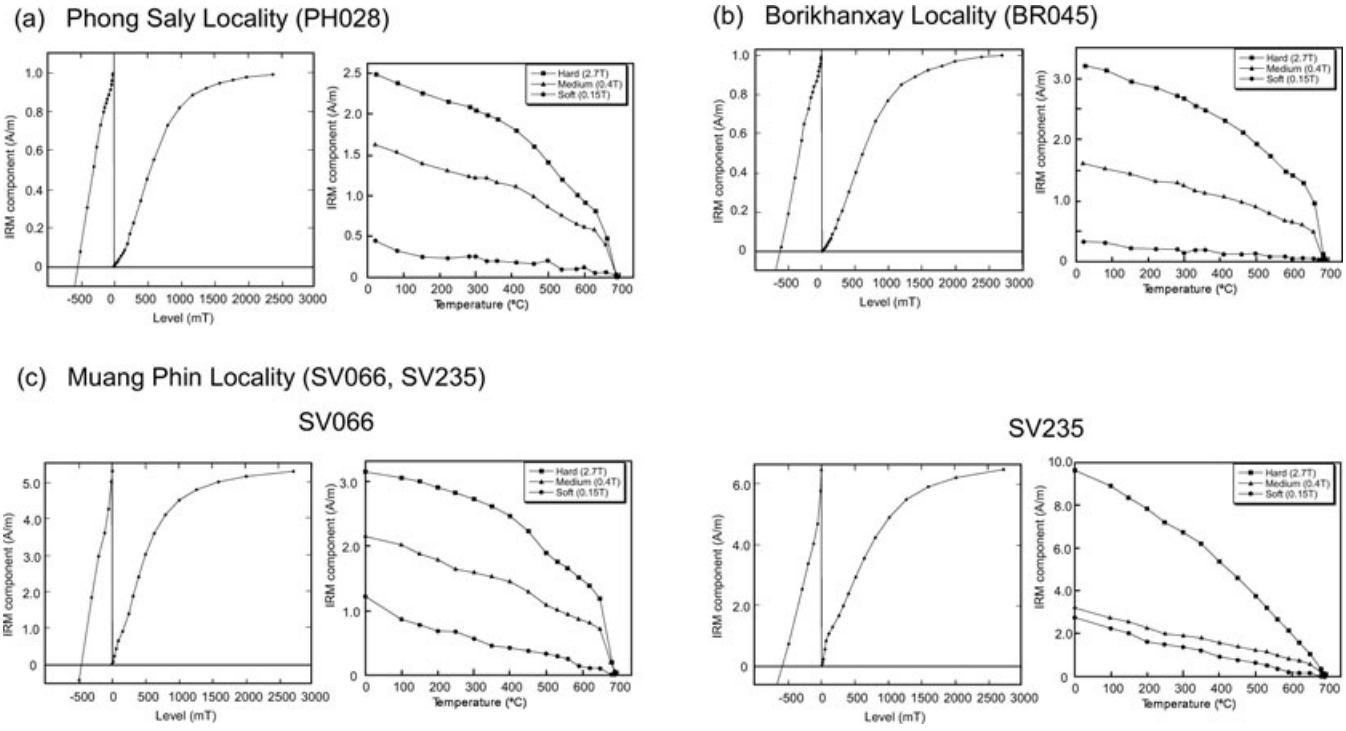


Figure 5. IRM acquisition and back-field demagnetization (Left box) and thermal demagnetization of the composite IRMs with applied DC fields of 2.7T, 0.4T and 0.12T along three perpendicular axes (right box) for representative specimens from (a) Phong Saly (PH028), (b) Borikhanxay (BR045) and (c) Mue Phin (SV066, SV235) localities. Even up to a maximum field of 2.7T, the IRM saturation level is not achieved. For all specimens, the thermal demagnetization of hard component indicates an unblocking temperature of hematite. The unblocking temperature of about 580 °C is observed in the soft component of two specimens from Muang Phin locality.

Yang & Besse (1993) and Charusiri *et al.* (2006) have reported palaeomagnetic results from the Jurassic to Cretaceous formations of the Khorat Basin, where an easterly-deflected declination of 27°–42° has been obtained. The Borikhanxay locality (Dec = 41.8°) is located in the northernmost part of the Khorat Basin, while the Muang Phin locality (Dec = 34.2°) is located in the easternmost part of this basin. Addition of these palaeomagnetic results with previously reported data allow us to extend the zone of easterly deflected declinations by 30 km towards north and 200 km towards east.

By using the available declination data, we have calculated rotational motion for different localities and then used them to document deformational features in the Shan-Thai and Indochina blocks. In this context, we have calculated an amount of rotation for 29 localities with respect to the SCB (Table 2). Using a compilation of highly reliable Cretaceous palaeomagnetic data sets from coastal area of the SCB and the Sichuan Basin (Yokoyama *et al.* 2001; Wang & Yang 2007); we have established a palaeomagnetic pole for the SCB (Table 2). By using a General Mapping Tools of Wessel & Smith (1991), contours of rotational angle are drawn. After assigning a null value to several localities along the margin of the Shan-Thai Block, a solution based on tension parameter $T_I = 0.4$ (Smith & Wessel 1990; Wessel & Smith 1991) is applied to depict first order approximation of rotation. We have constructed two separate contour maps by using all 29 data sets as well as the selected 18 data sets (which passed the fold test). Because of clear similarity between these two maps, a contour map with 29 data sets is chosen for further discussion (Fig. 6).

From this map, two visible rotational features are identified (Fig. 6): (1) the range of 10°clockwise rotation covers wide areas in the Shan-Thai and Indochina blocks and (2) nearly uniform clockwise rotation of 10° is observed in the Khorat Basin, while a variable magnitude of clockwise rotation (up to 105°) is observed in the Shan-Thai Block. Rotational motion and other aspects of deformation in the Shan Thai and Indochina blocks are given as following.

5.1 Rotational motion of the Shan-Thai and Indochina blocks as a composite unit

As a first order approximation, the above-mentioned rotational information implies that the Shan-Thai and Indochina blocks underwent a clockwise rotation as a composite unit by 10°. As evident from the contour map (Fig. 6), an area with 10° clockwise rotation covers about 76 per cent of the Shan-Thai Block and 30 per cent of the Indochina Block. Wide coverage of both these blocks by 10°contour implies that after the India–Asia collision these blocks underwent a rotational motion as a composite tectonic unit rather than independently.

This composite body dynamic model is supported by southward displacement of the Shan-Thai and Indochina blocks. We have calculated an amount of latitudinal displacement for all studied localities in the Shan-Thai and Indochina blocks with respect to the SCB (Table 2). For this purpose, data originated from the contour zone of 10° are used, and almost identical values of southward displacement is evaluated for the Shan-Thai (6.9° ± 1.1°) and Indochina (7.5° ± 6.8°) blocks. An arithmetic mean is computed by using a latitudinal difference between these localities (listed in Table 2) and its uncertainty is 95 per cent confidence limit which is calculated using the formula $t_{0.975}Sn^{-1/2}$, where $t_{0.975}$ is the Student's-t probability value of 0.975, S is the standard deviation and n is the

Table 2. The Cretaceous and Jurassic palaeomagnetic results available from Shan-Thai and Indochina blocks.

Locality name			Age	*N*	Fold test	Observed direction			VGP			Rotation	Latitudinal difference		Reference
	Lat. (°N)	Long. (°E)				Dec. (°)	Inc. (°)	α_{95} (°)	Lat. (°N)	Long. (°E)	A_{95} (°)	(°)	(°)	Relative to	
Shan-Thai block															
Yunlong	25.8	99.4	K2	20	Positive	40.2	49.9	3.9	54.6	171.3	4.4	29.6 ± 6.0	7.4 ± 4.8	A	Sato *et al.* (1999)
Yunlong	25.8	99.4	K2	29	Positive	38.3	50.7	3.4	56.7	170.1	4.0	27.7 ± 5.5	8.2 ± 4.5	A	Yang *et al.* (2001)
Xiaguan	25.6	100.2	K2	9	Positive	6.9	47.7	8.6	83.6	152.7	10.0	−3.8 ± 10.8	5.6 ± 8.6	A	Huang & Opdyke (1993)
Weishan	25.4	100.2	J3	5		7.3	25.3	10.4	76.3	250.0	10.4	−12.0 ± 10.6	6.0 ± 9.9	B	Huang & Opdyke (1993)
Yongping	25.5	99.5	K**l**	12	Positive	42.0	51.1	15.7	50.9	167.3	20.6	31.4 ± 20.7	8.8 ± 16.8	A	Funahara *et al.* (1993)
Jingdong	24.5	100.8	Kl-2	13	Positive	8.3	48.8	7.7	81.2	145.8	8.9	−2.3 ± 10.0	7.5 ± 7.8	A	Tanaka *et al.* (2008)
Luxi	24.3	98.4	J2	6		99.7	353.2	11.3	−0.5	116.6	12.2	81.0 ± 12.2	13.5 ± 11.1	B	Huang & Opdyke (1993)
Zhengyuan	24.1	101.1	Kl-2	7	Positive	61.8	46.1	8.1	34.7	172.7	8.1	51.2 ± 10.0	5.6 ± 7.2	A	Tanaka *et al.* (2008)
West Zhengyuan	24.1	101.1	Kl-2	4	Positive	324.2	−49.4	6.4	−25.7	135.2	7.7	133.6 ± 8.6	8.4 ± 6.9	A	Tanaka *et al.* (2008)
Jinggu	23.6	100.5	J2	10		83.3	36.8	5.4	14.0	173.6	4.2	64.0 ± 7.6	14.7 ± 6.5	B	Chen *et al.* (1995)
Jinggu	23.4	100.4	K**l**	3		84.4	39.6	17.8	13.6	171.5	-	73.9 ± 19.0	1.4 ± 10.7	A	Chen *et al.* (1995)
Jinggu	23.4	100.5	K2	7	Positive	295.8	−36.0	6.3	−13.9	161.3	-	105.3 ± 7.1	−1.1± 4.7	A	Chen *et al.* (1995)
Jinggu	23.4	100.9	K2	8		79.4	43.3	9.1	18.9	170.0	8.9	68.9 ± 10.6	4.1± 7.8	A	Huang & Opdyke (1993)
Pu'er	23.0	101.0	Kl-2	25	Positive	59.9	45.2	5.1	35.8	173.1	5.6	49.4 ± 6.7	6.0 ±5.5	A	Sato *et al.* (2007)
Mengla	21.6	101.4	K2	10		60.8	37.8	7.6	33.7	179.3	8.2	50.4 ± 8.4	1.7 ± 7.3	A	Huang & Opdyke (1993)
South Mengla	21.4	101.6	Kl-2	14	Positive	51.2	46.4	5.6	43.6	172.1	6.1	40.7 ± 9.9	8.5 ± 5.8	A	Tanaka *et al.* (2008)
Phong Saly	21.6	101.9	J3-K1	19	Positive	28.8	32.1	8.8	63.4	193.9	7.4	19.6 ± 10.4	−0.2 ± 6.7	A	This study
Kalaw	20.7	96.5	J-K	13	Positive	44.7	23.4	6.1	47.2	190.6	4.8	27.3 ± 7.9	9.6 ± 6.6	B	Richeter & Fuller (1996)
Nan	19.2	101.0	Jl-3	11	Positive	32.2	33.3	12.2	60.1	186.5	11.7	12.8 ± 12.9	16.3 ± 10.8	B	Aihara *et al.* (2007)
Indochina block															
Lai Chau	22.3	103.4	K3	5	Positive	12.2	40.1	4.7	78.7	188.0	5.1	1.5 ± 6.6	2.3 ± 5.2	A	Takemoto *et al.* (2005)
Yen Chau	21.0	104.4	K3	8	Positive	3.2	26.7	12.9	83.2	255.6	10.8	−7.4 ± 12.4	−5.1± 9.2	A	Takemoto *et al.* (2005)
Borikhanxay	18.5	103.8	J3-K1	18	Positive	42.1	46.9	7.9	50.7	169.7	8.7	31.7 ± 11.1	11.4 ± 7.7	A	This study
Amphoe Bung Kuan	18.2	103.9	K2	14		31.8	28.7	3.5	59.4	190.8	3.5	21.4 ± 5.4	−1.1 ± 4.3	A	Charusiri *et al.* (2006)
Nam Nao	16.5	103.0	K**l**	10		28.1	40.5	2.4	62.7	173.3	2.4	17.8 ± 5.0	8.5 ± 3.7	A	Yang & Besse (1993)
Nam Nao	16.5	103.0	J3	10		26.6	37.7	2.6	64.8	178.1	2.3	7.3 ± 6.4	20.5 ± 5.7	B	Yang & Besse (1993)
Muan Sakon Nakhon	16.5–17.2	102.5–104.1	K2	8		31.4	27.1	9.4	59.7	192.7	9.4	21.1 ± 9.5	−0.7 ± 8.2	A	Charusiri *et al.* (2006)
Muan Sakon Nakhon	16.5–17.2	102.5–104.1	K**l**	4	Positive	31.8	38.3	5.7	59.7	178.2	5.7	21.5 ±7.3	6.5 ± 5.6	A	Charusiri *et al.* (2006)
Muang Phin	16.5	106.0	J(Mid K)	23	Syn-folding	30.8	39.9	3.0	60.5	178.6	3.0	20.4±5.4	7.6±4.0	A	This study
				(18		29.5	39.9	3.4	61.6	178.3	3.4	19.1± 5.6	7.6± 4.2	A)	
Da Lat	10.4–12.5	105.0–109.4	J-K	21		14.5	33.3	6.3	74.2	171.1	5.9	4.3 ±7.4	7.8±5.7	A	Chi & Dorobek (2004)
(A) Cretaceous															
									76.3	172.6					
Nanjing	32.0	119.0	K	10					76.3	172.6	10.3				Kent *et al.* (1986)
Anhui	30.8	118.2	K	4					83.8	200.3	14.6				Gilder *et al.* (1999)
Zhejiang	29.0	119.7	K	10					83.2	247.9	10.6				Morinaga &Liu (2004)
Zhejiang	28.4	119.9	K	9					77.1	199.4	7.0				Morinaga & Liu (2004)
Fujian	26.4	117.8	K	8					76.3	197.1	12.3				Morinaga & Liu (2004)
Fujian	25.7	116.8	K	6					81.4	208.7	10.5				Morinaga & Liu (2004)
Fujian	25.1	116.4	K	5					85.5	167.8	9.9				Morinaga & Liu (2004)
Guangdong	24.1	115.8	K	7					81.9	215.6	6.1				Morinaga & Liu (2004)
Hong Kong	22.2	114.1	K	12					69.3	211.2	8.9				Li *et al.* (2005)
Mean									80.0	201.9	4.0				

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Table 2. (*Continued.*)

Locality name			Age	*N*	Fold test	Observed direction		VGP			Rotation	Latitudinal difference		Reference
	Lat. (°N)	Long. (°E)				Dec. (°)	Inc. (°)	α_{95} (°)	Lat. (°N)	Long. (°E)	A_{95} (°)	(°)	Relative to	
(B) Jurassic														
Wangchang	32.1	106.2	J3	6					66.6	236.4	7.6			Bai *et al.* (1998)
Bazhong	31.8	106.7	J3	11					67.4	236.0	7.0			Yokoyama *et al.* (2001)
Jiangyang	30.4	104.5	J3	24					61.3	222.7	4.2			Yokoyama *et al.* (2001)
Mean									64.3	231.2	6.7			

Notes: Lat. and Long., latitude and longitude. Age: J, Jurassic, J1, Lower Jurassic, J2, Middle Jurassic, J3, Upper Jurassic; K, Cretaceous, K1, Lower Cretaceous; K2, Upper Cretaceous. N, number of sites used for palaeomagnetic statistics. Dec. and Inc., declination and inclination. α_{95} and A_{95} are radii of cone of 95 per cent confidence about the mean direction and the VGP, respectively. Rotational and latitudinal differences are evaluated by comparing the observed palaeomagnetic declinations with those expected from the Cretaceous (A) and Jurassic (B) palaeomagnetic poles of the SCB. Uncertainty in rotational and latitudinal displacement is calculated after Demarest (1983).

number of data. We, therefore, conclude that the Shan-Thai and Indochina blocks behaved as a rigid body since the Cretaceous and experienced a clockwise rotation of about 10° together with southward displacement of 7.1° ± 2.2°. Clockwise rotation together with southeastward extrusion of these two blocks as a single tectonic unit has been previously discussed by Tapponnier *et al.* (1982), Yang & Besse (1993) and Replumaz & Tapponnier (2003) in their indentation models. Our conclusion in favour of rigid body model thus provides rationale for these assumptions.

According to their 4-D evolution model reconstructed from seismic tomography, Replumaz *et al.* (2004) have placed this event of rotation as a composite unit prior to the commencement of southward displacement. In several other studies, southward displacement of the Shan-Thai and Indochina blocks has been attributed to sinistral motion along the Ailao Shan-Red River fault zone, which was tectonically active between 32 and 17 Ma (Leloup *et al.* 1995, 2001; Wang & Burchfiel 2000; Wang *et al.* 2001a; Gilley *et al.* 2003). This type of interpretation thus suggests that the composite unit (composed of Shan-Thai and Indochina blocks) experienced a clockwise rotation of about 10° during the early stage of indentation (India in to Asia). Comparison with previous studies allows us to suggest that this clockwise rotation as a composite unit took place prior to 32 Ma. That was a time when this combined unit achieved a squeeze out of gap between the Qiangtang and Lhasa blocks, which presently form part of the Tibetan Plateau (Otofuji *et al.* 2007).

5.2 A rigid Indochina Block versus deformable Shan-Thai Block

Features of clockwise rotation in the Indochina Block are quite different from those in the Shan-Thai Block. The Indochina Block experienced a uniform clockwise rotation of 10°, whereas variable internal deformation associated with clockwise rotation (between −5° and 105°) is estimated for the Shan-Thai Block. An internal deformation of the Shan-Thai Block has been ascribed to the formation of sinusoidal shaped Chongshan-Lancang-Chiang Mai belt during 32 to 27Ma as a result of north–south compressive regime (Tanaka *et al.* 2008). This extremely large amount of clockwise rotation is partly ascribed to internal displacement of microterranes within the Shan Thai Block as a result of Pliocene–Quaternary activation along the network of faults (Lacassin *et al.* 1998). Based on these observations it is suggested that the Indochina block behaved as a rigid body, while the Shan Thai Block behaved as a mechanically weak medium.

This contrast in internal deformation between the Indochina and Shan-Thai blocks has been observed by recent GPS measurements. Based on the horizontal GPS data it has been reported that the Shan-Thai Block is divided into three subblocks, that is, the Baoshan, Lincang and central Yunnan subblocks in displacement-direction order (Shen *et al.* 2005; Gan *et al.* 2007). These blocks are moving southwestward, southward and southeastward, respectively, with respect to stable Eurasia at fast speed (Gan *et al.* 2007). On the other hand, a uniform eastward displacement of about 10 mm yr^{-1} has been estimated for Indochina Block with respect to Eurasia (Calais *et al.* 2006). Presently, the Indochina Block is considered as a part of Sundaland (Simons *et al.* 2007).

Lack of internal deformation in the Indochina Block is also reflected from the physical nature of its lithosphere. Based on *P*-wave variation in the upper mantle, a high velocity anomalous zone has been detected at a depth of 140 km to 200 km beneath the Khorat Basin (Li *et al.* 2006). This type of behaviour implies the presence of

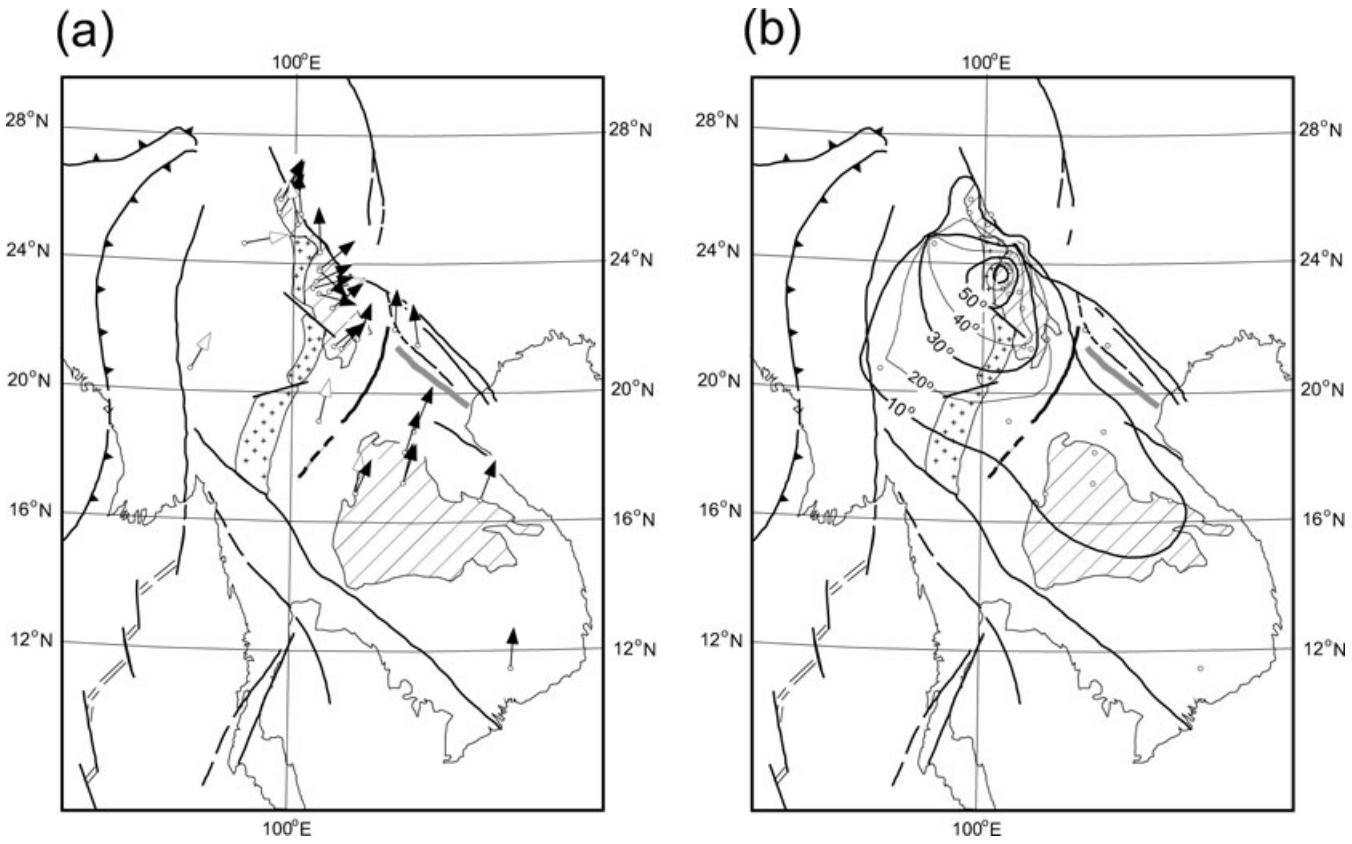


Figure 6. (a) Magnitude of palaeomagnetic rotation estimated for each locality with respect to SCB is shown by arrows. Deflection of arrows from north indicates an amount of rotation. (Jurassic: open arrow, Cretaceous: solid arrow). (b) Contours showing an amount of rotation in the Shan-Thai and Indochina blocks. These contours are plotted at 10 degrees interval by inputting the available rotational data from 29 localities (as listed in Table 2) by using the General Mapping Tools of Wessel & Smith (1991). A null rotation is assumed for several areas, which fall in a boundary area between the Shan-Thai and Indochina blocks.

continental roots beneath the Khorat Basin, which may be mechanically strong enough to resist the impact of tectonic deformation. In contrast, *P*- and *S*-wave velocity tomography indicates an absence of continental roots beneath the Shan-Thai Block (Lebedev & Nolet 2003; Li *et al.* 2006). The high–resolution *P*-wave tomography of the Shan-Thai Block further indicates a prominent low velocity zone up to 300 km deep under the Tenchong volcano (Huang & Zhao 2006). A preliminary 3-D *S*-wave velocity structure also suggests a presence of low velocity anomaly (sub-Moho) beneath the Tenchong volcano (Hunag *et al.* 2003). These low *P*- and *S*-wave velocity zones can be linked to the presence of hot asthenosphere beneath the Shan-Thai Block, which may be the reason for weak mechanical strength of the overlying crust. Based on these points, we thus conclude that thick and strong lithosphere under the Indochina Block allows it to behave as a rigid block without any internal deformation, while an absence of roots beneath the Shan-Thai Block gave rise to an extensive internal deformation in its lithosphere.

6 CONCLUSION

(1) An easterly deflected declination ($D = 28.8°$–$42.1°$) is observed in the Upper Jurassic to Middle Cretaceous red beds of Phong Saly (21.6°N, 101.9°E), Borikhanxay (18.5°N, 103.8°E) and Muang Phin (16.5°N, 106.0°E) localities, located in the Indochina Peninsula.

(2) By using an integrated palaeomagnetic database of previous and present studies, tectonics of the Shan-Thai and Indochina blocks is described by two distinct phenomenon, (1) after an initial indentation of India in to Asia, the Shan-Thai and Indochina blocks behaved as a composite tectonic unit with clockwise rotation of about 10°, (2) following this, nearly uniform clockwise rotation of about 10° is observed within the Khorat Basin of Indochina Block, whereas variable amount of clockwise rotation (between −5° and 105°) is observed within the Shan-Thai Block.

(3) Comparison of our palaeomagnetic results with seismic tomographic images suggests that thick and rigid lithosphere beneath the Indochina Block saved it from internal deformation, while an absence of strong roots under the Shan-Thai Block paved a way for extensive internal deformation of its ancient lithosphere.

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